ENR 3
ENR 3 ATS ROUTES
ENR 3.1 LOWER ATS ROUTES

Route designator ***RCP Type*** ***Name of significant points*** ***Coordinates***	***Route*** ***MAG*** ***Track*** ***MAG*** ***DIST***	***Upper limit*** ***Lower limit*** ***Classification***	***Lateral limits*** ***NM***	***Direction of cruising levels***		***Remarks*** ***Controlling unit*** ***Frequency*** ***SATVOICE*** ***Logon address***
				Odd	***Even***	
1	***2***	***3***	***4***	***5***		***6***
A402						
▲CAPE TOWN INTERNATIONAL AIRPORT (CTV) VOR/DME 335810.82S 0183618.19E						
	126° 306° 44 NM	FL 245 FL 145 Class A	10	↓	↑	
▲CALEDON (CDV) VOR 340646.68S 0192840.39E						
	113° 293° 145 NM	FL 245 FL 145 Class A	10	↓	↑	
▲GEORGE AIRPORT (GRV) VOR/DME 340026.46S 0222233.46E						
	117° 296° 161 NM	FL 245 FL 145 Class A	10	↓	↑	
▲CHIEF DAWID STUURMAN AIRPORT (PEV) VOR/DME 335933.01S 0253620.75E						
	091° 270° 125 NM	FL 245 FL 145 Class A	10	↓	↑	
▲KING PHALO AIRPORT (ELV) VOR/DME 330213.16S 0274923.36E						
	073° 251° 121 NM	FL 245 FL 145 Class A	10	↓	↑	
▲PORT ST JOHNS (PJV) VOR 313631.96S 0293111.49E						FAJA/FACA FIR BDRY

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	062° 240° 96 NM	FL 245 FL 145 Class A	10	↓	↑	
▲OKTAN 301758.32S 0303453.34E						
	060° 239° 50 NM	FL 245 FL 145 Class A	10	↓	↑	
▲KING SHAKA (TGV) VOR/DME 293640.20S 0310729.03E						
	117° 297° 50 NM	FL 245 FL 145 Class A	10	↓	↑	
▲EVIDO 293844.82S 0320448.90E						
	094° 273° 59 NM	FL 245 FL 145 Class A	10	↓	↑	
▲ESRET 291639.66S 0330805.46E						
	095° 275° 72 NM	FL 245 FL 145 Class A	10	↓	↑	
▲APLUK 285213.00S 0342532.00E						FAJA/FAJO FIR BDRY
	095° 274° 85 NM	FL 245 FL 145 Class A	10	↓	↑	
▲ETMOS 282224.00S 0355618.00E						FAJO/FQBE FIR BDRY

A405						
▲GREEFSWALD (GWV) VOR 221444.60S 0292437.21E						FAJA/FVHF FIR BDRY
	215° 038° 124 NM	FL 245 FL 145 Class A	10	↑	↓	
▲GABNA 241000.00S 0283400.00E						
	219° 041° 99 NM	FL 245 FL 145 Class A	10	↑	↓	

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲UTRUK 254032.98S 0274957.29E						
	207° 028° 51 NM	FL 245 FL 145 Class A	10	↑	↓	
▲GRASMERE (GAV) VOR 263053.63S 0274035.17E						
	248° 069° 135 NM	FL 245 FL 145 Class A	10	↑	↓	
▲EKBEX 280039.10S 0254723.96E						
	249° 069° 28 NM	FL 245 FL 145 Class A	10	↑	↓	
▲APDUR 281919.65S 0252312.80E						
	249° 070° 44 NM	FL 245 FL 145 Class A	10	↑	↓	
▲KIMBERLEY AIRPORT (KYV) VOR/DME 284759.82S 0244537.32E						
	229° 052° 162 NM	FL 245 FL 145 Class A	10	↑	↓	
▲EGRUP 311005.95S 0231539.78E						FAJA/FACA FIR BDRY
	233° 053° 16 NM	FL 245 FL 145 Class A	10	↑	↓	
▲VICTORIA WEST (VWV) VOR 312349.00S 0230632.00E						
	269° 088° 142 NM	FL 245 FL 145 Class A	10	↑	↓	
▲SUTHERLAND (SLV) VOR 322700.00S 0203721.00E						
	255° 076° 77 NM	FL 245 FL 145 Class A	10	↑	↓	
▲CERES (CSV) VOR/DME 331508.27S 0192617.22E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	248° 069° 60 NM	FL 245 FL 145 Class A	10	↑	↓	
▲CAPE TOWN INTERNATIONAL AIRPORT (CTV) VOR/DME 335810.82S 0183618.19E						

B529						
▲KING SHAKA (TGV) VOR/DME 293640.20S 0310729.03E						
	070° 250° 50 NM	FL 245 FL 145 Class A FL 145 FL 075 Class C	10	↓	↑	
▲ITMIL 290151.05S 0314838.40E						
	064° 244° 23 NM	FL 245 FL 145 Class A FL 145 FL 075 Class C	10	↓	↑	
▲RICHARDS BAY (RBV) VOR/DME 284426.49S 0320536.91E						

B534						
▲WITBANK (WIV) VOR 254946.29S 0291141.13E						
	128° 309° 91 NM	FL 245 FL 145 Class A	10	↓	↑	
▲DUKRA 262024.43S 0304711.57						FAJA/FDMS TMA BDRY

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	149° 330° 27 NM	FL 245 FL 145 Class A	10	↓	↑	
▲MUVAL 263728.00S 0310957.00E						

G465						
▲OKTEL 280753.81S 0150000.00E						FAJA/FAJO FIR BDRY
	127° 306° 85 NM	FL 245 FL 200 Class A FL 195 FL 145 Class C	10	↓	↑	
▲ALEXANDER BAY (ABV) VOR/DME 283414.35S 0163201.80E						
	108° 286° 236 NM	FL 245 FL 200 Class A FL 195 FL 145 Class C	10	↓	↑	
▲IBVIT 282453.62S 0210000.44E						
	106° 286° 14 NM	FL 195 FL 145 Class G	10	↓	↑	Class G between IBVIT and XUGUV until ATS Surveillance is established between IBVIT and XUGUV
▲UPINGTON AIRPORT (UPV) VOR/DME 282406.31S 0211537.48E						
	083° 263° 103 NM	FL 195 FL 145 Class G	10	↓	↑	Class G between IBVIT and XUGUV until ATS Surveillance is established between IBVIT and XUGUV
▲SISHEN (SSV) VOR/DME 273907.83S 0230010.99E						

Route designator ***RCP Type*** ***Name of significant points*** ***Coordinates***	***Route*** ***MAG*** ***Track*** ***MAG*** ***DIST***	***Upper limit*** ***Lower limit*** ***Classification***	***Lateral limits*** ***NM***	***Direction of cruising levels***		***Remarks*** ***Controlling unit*** ***Frequency*** ***SATVOICE*** ***Logon address***
				Odd	***Even***	
1	***2***	***3***	***4***	***5***		***6***
	086° 266° 60 NM	FL 195 FL 145 Class G	10	↓	↑	Class G between IBVIT and XUGUV until ATS Surveillance is established between IBVIT and XUGUV
▲XUGUV 271613.43S 0240233.27E						
	087° 266° 223 NM	FL 245 FL 200 Class A FL 195 FL 145 Class C	10	↓	↑	
▲LANSERIA AIRPORT (LIV) VOR/DME 255656.14S 0275448.86E						
	102° 281° 70 NM	FL 245 FL 200 Class A FL 195 FL 145 Class C	10	↓	↑	
▲WITBANK (WIV) VOR 254946.29S 0291141.13E						
	059° 240° 95 NM	FL 245 FL 200 Class A FL 195 FL 145 Class G	10	↓	↑	
▲ANVIT 244000.00S 0302300.00E						

G467						
▲KRUGER MPUMALANGA (PKV) VOR 252251.69S 0310640.05E						
	069° 249° 20 NM	FL 245 FL 200 Class A	10	↓	↑	

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲DUSKO 251028.74S 0312401.70E						
	069° 248° 42 NM	FL 245 FL 200 Class A	10	↓	↑	
▲TEVAP 244430.00S 0320006.00E						FAJA/FQBE FIR BDRY

G653						
▲ETOSA 244250.40S 0260312.98E						FAJA/FGBR FIR BDRY
	137° 318° 36 NM	FL 195 FL 145 Class A	10	↓	↑	Bidirectional
▲NESEK 250154.00S 0263700.00E						
	137° 317° 77 NM	FL 195 FL 145 Class A	10	↓	↑	Bidirectional
▲UTRUK 254032.98S 0274957.29E						
	161° 341° 36 NM	FL 195 FL 145 Class A	10	↓	↑	Bidirectional
▲O R TAMBO INTERNATIONAL AIRPORT (JSV) VOR/DME 260925.63S 0281351.70E						
	088° 267° 56 NM	FL 195 FL 145 Class A	10	↓	↑	Bidirectional
▲WITBANK (WIV) VOR 254946.29S 0291141.13E						
	111° 81 NM	FL 195 FL 145 Class A	10	↓		Directional (Eastbound only)
▲OKRIP 255258.16S 0304133.91E						
	111° 25 NM	FL 195 FL 145 Class A	10	↓		Directional (Eastbound only)
▲ESRAS 255346.00S 0310914.02E						

Route designator ***RCP Type*** ***Name of significant points*** ***Coordinates***	***Route*** ***MAG*** ***Track*** ***MAG*** ***DIST***	***Upper limit*** ***Lower limit*** ***Classification***	***Lateral limits*** ***NM***	***Direction of cruising levels***		***Remarks*** ***Controlling unit*** ***Frequency*** ***SATVOICE*** ***Logon address***
				Odd	***Even***	
1	***2***	***3***	***4***	***5***		***6***
	111° 31 NM	FL 195 FL 145 Class A	10	↓		Directional (Eastbound only)
▲ESRUM 255439.72S 0314330.38E						
	111° 13 NM	FL 195 FL 145 Class A	10	↓		Directional (Eastbound only)
▲ORNAD 255500.00S 0315800.00E						FAJA/FQBE FIR BDRY

G655						
▲MASERU (MZV) VOR 293001.00S 0273226.00E						
	036° 212° 172 NM	FL 245 FL 145 Class A	10	↓	↑	
▲HEIDELBERG (HGV) VOR 264148.50S 0281659.39E						
	357° 176° 66 NM	FL 245 FL 145 Class A	10	↑	↓	
▲UTRUK 254032.98S 0274957.29E						
	026° 204° 93 NM	FL 245 FL 145 Class A	10	↓	↑	
▲ODKER 240830.00S 0280500.00E						
	023° 200° 92 NM	FL 245 FL 145 Class A	10	↓	↑	
▲TAVLA 223724.00S 0281736.00E						FAJA/FBGR FIR BDRY

G745		
▲O R TAMBO INTERNATIONAL AIRPORT (JSV) VOR/DME 260925.63S 0281351.70E		

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	268° 56 NM	FL 245 FL 145 Class A	10		↑	Directional (Westbound only)
▲WITBANK (WIV) VOR 254946.29S 0291141.13E						
	273° 77 NM	FL 245 FL 145 Class A	10		↑	Directional (Westbound only)
▲DODOS 253032.17S 0303437.62E						
	273° 30 NM	FL 245 FL 145 Class A	10		↑	Directional (Westbound only)
▲KRUGER MPUMALANGA (PKV) VOR 252251.69S 0310640.05E						
	308° 30 NM	FL 195 FL 145 Class A	10		↑	Directional (Westbound only)
▲GAGNO 253309.23S 0313749.16E						
	308° 21 NM	FL 195 FL 145 Class A	10		↑	Directional (Westbound only)
▲ANVAK 254020.71S 0315950.23E						
	309° 35 NM	FL 145 FL 095 Class C	10		↑	Directional (Westbound only)
▲MAPUTO (VMA) VOR 255208.00S 0323616.00E						FAJA/FQBE FIR BDRY

G853						
▲KING SHAKA (TGV) VOR/DME 293640.20S 0310729.03E						
	291° 111° 38 NM	FL 245 FL 145 Class A	10	↑	↓	
▲PIETERMARITZBURG (PMV) VOR/DME 293852.13S 0302401.28E						

Route designator ***RCP Type*** ***Name of significant points*** ***Coordinates***	***Route*** ***MAG*** ***Track*** ***MAG*** ***DIST***	***Upper limit*** ***Lower limit*** ***Classification***	***Lateral limits*** ***NM***	***Direction of cruising levels***		***Remarks*** ***Controlling unit*** ***Frequency*** ***SATVOICE*** ***Logon address***
				Odd	***Even***	
1	***2***	***3***	***4***	***5***		***6***
	353° 172° 73 NM	FL 245 FL 145 Class A	10	↑	↓	
▲LADYSMITH (LYV) VOR/DME 283616.21S 0294140.32E						
	349° 167° 60 NM	FL 245 FL 145 Class A	10	↑	↓	
▲WARDEN (WRV) VOR 274629.26S 0290410.10E						
	348° 166° 77 NM	FL 245 FL 145 Class A	10	↑	↓	
▲HEIDELBERG (HGV) VOR 264148.50S 0281659.39E						
	357° 176° 66 NM	FL 245 FL 145 Class A	10	↑	↓	
▲UTRUK 254032.98S 0274957.29E						
	341° 159° 84 NM	FL 245 FL 145 Class A	10	↑	↓	
▲ORKAN 243230.00S 0265500.00E						
	338° 158° 16 NM	FL 245 FL 145 Class A	10	↑	↓	
▲RUDAS 241924.00S 0264406.00E						FAJA/FBGR FIR BDRY
R987						
▲ERDAS 332450.10S 0191112.85E						
	022° 199° 122 NM	FL 245 FL 145 Class A	10	↓	↑	
▲NIEUWOUDTVILLE (NVV) VOR 312248.00S 0190416.00E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	009° 187° 117 NM	FL 245 FL 145 Class A	10	↓	↑	
▲AGGENEYS (AGV) VOR/DME 292851.67S 0183344.97E						
	013° 193° 36 NM	FL 245 FL 145 Class A	10	↓	↑	
▲IBRUN 285324.75S 0182900.19E						
	013° 191° 84 NM	FL 245 FL 145 Class A	10	↓	↑	
▲KEBAT 273000.00S 0181800.00E						FAJA/FYWF FIR BDRY

W64						
▲KIMBERLEY AIRPORT (KYV) VOR/DME 284759.82S 0244537.32E						
	085° 265° 45 NM	FL 145 FL 105 Class C FL 195 FL 145 Class A	10	↓	↑	
▲APMAK 282849.76S 0253155.21E						
	085° 265° 66 NM	FL 195 FL 145 Class A	10	↓	↑	
▲WELKOM (WMV) VOR/DME 280005.74S 0263940.13E						

W66		
▲GEORGE AIRPORT (GRV) VOR/DME 340026.46S 0222233.46E		

Route designator ***RCP Type*** ***Name of significant points*** ***Coordinates***	***Route*** ***MAG*** ***Track*** ***MAG*** ***DIST***	***Upper limit*** ***Lower limit*** ***Classification***	***Lateral limits*** ***NM***	***Direction of cruising levels***		***Remarks*** ***Controlling unit*** ***Frequency*** ***SATVOICE*** ***Logon address***
				Odd	***Even***	
1	***2***	***3***	***4***	***5***		***6***
	055° 232° 187 NM	FL 245 FL 145 Class A	10	↓	↑	
▲OKSOP 311556.12S 0240832.74E						FAJA/FACA FIR BDRY
	052° 231° 61 NM	FL 245 FL 145 Class A	10	↓	↑	
▲PETRUSVILLE (PVV) VOR 302158.41S 0244142.06E						

W81						
▲CHIEF DAWID STUURMAN AIRPORT (PEV) VOR/DME 335933.01S 0253620.75E						
	032° 212° 50 NM	FL 245 FL 145 Class A	10	↓	↑	
▲EGTIM 330937.71S 0254125.96E						
	032° 210° 93 NM	FL 245 FL 145 Class A	10	↓	↑	
▲HOFMEYER (HMV) DVOR 313621.18S 0255038.40E						
	034° 213° 41 NM	FL 245 FL 145 Class A	10	↓	↑	
▲UTSEM 305604.65S 0255807.95E						FAJA/FACA FIR BDRY
	033° 212° 66 NM	FL 245 FL 145 Class A	10	↓	↑	
▲GEDOL 295037.49S 0261004.18E						
	032° 211° 45 NM	FL 245 FL 145 Class A	10	↓	↑	

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲BRAM FISCHER INTL AIRPORT (BLV) VOR/DME 290602.38S 0261802.31E						
W95						
▲VICTORIA WEST (VWV) VOR 312349.00S 0230632.00E						
	074° 254° 19 NM	FL 245 FL 145 Class A	10	↓	↑	
▲OKBIL 311244.26S 0232351.31E						FAJA/FACA FIR BDRY
	074° 254° 84 NM	FL 245 FL 145 Class A	10	↓	↑	
▲PETRUSVILLE (PVV) VOR 302158.41S 0244142.06E						
	069° 249° 113 NM	FL 245 FL 145 Class A	10	↓	↑	
▲BRAM FISCHER INTL AIRPORT (BLV) VOR/DME 290602.38S 0261802.31E						
	037° 216° 45 NM	FL 245 FL 145 Class A	10	↓	↑	
▲AVUBU 282242.79S 0263218.00E						
	036° 216° 23 NM	FL 245 FL 145 Class A	10	↓	↑	
▲WELKOM (WMV) VOR/DME 280005.74S 0263940.13E						
	051° 229° 104 NM	FL 245 FL 145 Class A	10	↓	↑	
▲GRASMERE (GAV) VOR 263053.63S 0274035.17E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
R987F						
▲NIEUWOUDTVILLE (NVV) VOR 312248.00S 0190416.00E						
	009° 187° 117 NM	FL 195 FL 095 Class G	10	↓	↑	
▲AGGENEYS (AGV) VOR/DME 292851.67S 0183344.97E						
	013° 193° 36 NM	FL 195 FL 095 Class G	10	↓	↑	
▲IBRUN 285324.75S 0182900.19E						
	013° 191° 84 NM	FL 195 FL 095 Class G	10	↓	↑	
▲KEBAT 273000.00S 0181800.00E						FAJA/FYWF FIR BDRY